Dominik Hommer

AI Security and Applied AI Research | Co-Founder, Casdo Labs GmbH

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AI researcher and engineer working on efficient on-device AI security, computer vision and multimodal learning. Co-founder of Casdo Labs, developing Patronus Protect and publishing AI-security models on Hugging Face. B.Sc. graduate with international research experience and a first-author journal article accepted for publication.

Research and Engineering Experience

Co-Founder | Casdo Labs GmbH / Patronus Protect

Germany | 2026 - Present

- Developing an on-device AI firewall for prompt-injection detection, sensitive-data protection and monitoring of AI agents and tool activity across macOS and Windows.
- Working on provider-independent AI-traffic detection using network, protocol, application and system context, with local inspection and policy enforcement.
- Published AI-security models and deployment variants under patronus-studio on Hugging Face: 18 public repositories, 10,000+ monthly downloads and 40,000+ total downloads (13 Oct 2026 snapshot: 12,513 monthly; 49,007 all-time).
- Developing Patronus Ark, an open-source security engine combining native rules, lightweight classifiers and quantized transformer inference in a layered Rust/Python architecture.

Mitacs Globalink Research Internship | VisLang-Seg

Edmonton, Canada | May - Jul 2026

- Developed a resource-efficient, two-stage cancer-cell image-segmentation architecture combining lightweight preprocessing and candidate selection with Vision Transformers and LLM-based multimodal segmentation.

Bachelor Thesis | Geometry-Aware Table Segmentation

OTH Regensburg | Oct 2025 - Apr 2026

- Designed a geometry-aware loss for ViT-based table-cell segmentation with axis-alignment, morphological closing and spur-suppression regularization.
- Evaluated on PubTables-1M against a BCE-Dice baseline; reported gains of up to 9 percentage points in region-level detection with comparable pixel-level performance and no added inference-time overhead.

Document Digitization | DocDig

2024 - 2026

- Developed deep-learning and computer-vision methods to extract structured data from handwritten Swiss ornithological tables; presented a research poster at a scientific conference in Erfurt.

Publication

Dominik Hommer (first author). Geometry-Aware Losses for Structurally Consistent Table Segmentation. Array, Elsevier. Accepted 11 Oct 2026; in production. Derived from bachelor thesis.

Education

Master’s Program in Mathematics for Business and Industry | OTH Regensburg

Germany | Currently enrolled

B.Sc. Artificial Intelligence and Data Science | OTH Regensburg

Germany | 2022 - Apr 2026

Graduated 1 Apr 2026. Overall grade: 1.8 (German scale, 1.0 best); 210 ECTS. Coursework includes machine learning, deep learning, optimization, statistics and computer science.

Selected Open Models

Wolf Defender and Wolf Defender Small

Prompt-injection detection

- Multilingual ModernBERT/mmBERT classifiers with adversarial augmentation and hard-negative training. Small v2 reports 95.21% F1 on Qualifire (5,000 examples); available as FP32, FP16 and INT8/INT4 ONNX exports, including a 96.30 MB variant.

Lion Warden

Multi-task AI security

- Shared encoder with seven security heads for injection, threat type, sensitive documents, request routing and tool type, action and data-flow risk; masked multi-task training. Model card reports mean head F1 of 0.937 on its held-out test sets.

Wolf Defender Jev

Context-aware security decisions

- 151M-parameter decision model taking agent context, a security question and candidate answers; returns calibrated probabilities rather than generated text. Published with support for 15 languages and local CPU inference.

Additional releases cover document classification (Orca Sonar), tool risk (Husky), intent routing (Panther Read), shell-command risk (Cat Shell Risk Commander), and quantized GLiNER entity extraction. Benchmarks above are developer-reported model-card results.

Industry and Teaching Experience

Working Student | trinnovative GmbH

Regensburg | Apr 2025 - Apr 2026

- Worked on data science and backend development.

Programming Tutor | OTH Regensburg CodeClinic

Regensburg | Oct 2025 - Feb 2026

- Supported students with programming coursework and debugging.

Data Science and AI Internship | Audi AG

Ingolstadt | Oct 2024 - Mar 2025

- Developed unsupervised driving-style recognition using autoencoders, time-series methods and clustering; built document-grounded LLM assistants and analyzed automotive error sequences.

Working Student | AVL Software and Functions GmbH

Regensburg | May - Aug 2024

- Developed LLM-based and classical autocompletion using N-gram and Trie methods for a software-engineering web application; collaborated with international teams.

Mechatronics Apprenticeship | Siemens AG

Amberg | Sep 2018 - Aug 2021

- Worked with industrial automation and production systems; qualification awarded with special recognition and training shortened for performance.

Technical Skills and Recognition

Programming: Python, C, C++, SQL. ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, ONNX Runtime, NumPy, Pandas. Data and systems: PostgreSQL, PySpark, Databricks, backend development, Git, CI/CD.

Methods: transformer fine-tuning, multi-task learning, model evaluation, quantization, computer vision, OCR, multimodal learning, clustering and time-series analysis. Languages: German (native), English (fluent).

German Abitur Prize in Physics (2022).